Distributed Systems: Concurrency Control and Locking

1. Concurrency Control and Serializability
In multi-user database systems, concurrency control protocols guarantee that concurrent transaction executions are conflict

2. Two-Phase Locking Protocol (2PL)
The Two-Phase Locking (2PL) protocol enforces serializability through two distinct execution phases:
- Growing Phase: The transaction may acquire shared (read) or exclusive (write) locks on database items, but cannot releas
- Shrinking Phase: The transaction may release acquired locks, but is strictly prohibited from acquiring any new locks.
Theorem: If every transaction in a schedule follows 2PL, the resulting schedule is guaranteed to be conflict-serializable.

Consensus Protocols and Leader Election

1. Raft Consensus Protocol Communication
Raft achieves distributed consensus by electing a distinguished leader node. The protocol follows a sequence of messages
Step 1: The client sends a state machine command to the Leader node via RPC.
Step 2: The Leader node appends the entry to its local write-ahead log.
Step 3: The Leader sends AppendEntries RPC requests to all Follower nodes in the cluster.
Step 4: Followers process the log entry and respond with an acknowledgment RPC back to the Leader.
Step 5: Once a majority quorum responds, the Leader commits the entry and replies to the client.

2. Evolution of Consensus Protocols
Historical evolution and timeline of distributed fault tolerance:
In 1989, Paxos was formulated by Leslie Lamport. In 2014, Ongaro and Ousterhout developed Raft for understandability.